STATEMENT OF WORK (SOW) & MASTER SERVICES RATE CARD

Turnkey Rollout, Site Engineering, Telecom Integration, and 24/7 Managed Operations & Maintenance (O&M)

Author: OLIVER TUNGOL, PMP, PECE | Version: 2.1 (Reconciled Services SOW & Commercial Rate Card) | Release Date: 26 October 2026 | Audited Tower Database: 13,810 Records (13,799 Unique IDs / 13,629 Coordinates) | Baseline FX: 1 USD = 55.00 PHP

EXPLICIT GOVERNANCE & COMMERCIAL RECONCILIATION PRINCIPLES

1. Primary Operational Baseline (TowerCo-Led Scope on 13,810 Audited Tower Locations): The primary operational objective of the Operator is to utilize Independent Tower Companies (**TowerCos**) under Master Lease Agreements (MLA) — either through **Tier 1 (Existing Co-Location on the audited master database of 13,810 registry records / 13,799 unique IDs / 13,629 coordinates, 89.7% / 897 Sites)** or **Tier 2 (TowerCo Build-to-Suit / B2S, nominal 103 Sites)**. Under this primary baseline, **all Site Acquisition (SAQ) work** — including candidate hunting, landlord negotiations, title due diligence, lease execution, Barangay clearances, LGU building permits, CAAP clearances, and utility power applications — is **STRICTLY THE SCOPE AND FINANCIAL RESPONSIBILITY OF THE TOWERCO, NOT THE VENDOR**.

2. Structural Engineering & Permitting Ownership Boundary: Under MLA, for Tier 1 Co-Location and Tier 2 B2S, formal 3D structural analysis, TIA-222-H wind simulation (250–315 km/h site-specific basis), foundation capacity verification, and licensed Philippine Professional Civil/Structural Engineer (PE) wet-seal certification **STRICTLY BELONG TO THE TOWERCO** (per Appendix D §1.3 and Bid Bulletin 01 Huawei Item 13). Vendor scope is strictly limited to physical auditing, compiling active equipment loading specifications (weight, EPA, mounting brackets), and verifying the antenna mount interface. Vendor structural PE endorsement applies *strictly* to Tier 3 Operator Own-Build.

3. Standby SAQ Contingency Rate Card (10.3% Evaluation Basket): The unit rates requested under Category 1 (Items SAQ-01 to SAQ-08) serve **STRICTLY AS A STANDBY / CONTINGENCY CALL-OFF RATE CARD** ('Operator Self-Acquisition / Own-Build Option'). To allow fair commercial evaluation without inflating baseline contract commitments, SAQ lines are evaluated against an explicit **Standby Contingency Evaluation Basket of 103 Sites (Phase 1) / 515 Sites (Program)**, representing the 10.3% Own-Build ratio. Billing is strictly triggered upon formal issuance of Operator Work Orders for authorized self-build sites. Bidders shall NOT embed SAQ risk premiums into baseline turnkey hardware or installation pricing.

4. Annualized & 3-Year Contractual O&M Evaluation (No Double-Counting): Recurring 24/7 Managed Services are evaluated under standardized multi-year periods: Monthly retainers × 12 months; Quarterly PMs × 4 quarters; Semi-Annual PMs × 2 semi-annuals. Category B defines both an **Annualized 1-Year O&M Baseline** and a **3-Year Managed O&M Contractual Baseline (36 Months)** matching RFP Lot 6. Retainer-included items (FLM-02 to 04, SPM-02 to 03) evaluate at **$0.00** to guarantee zero double-charging.

1. PROGRAM OVERVIEW & SITING ARCHITECTURE

The 5G Fixed Wireless Access network utilizes 3GPP Band n50 with an authorized operational passband of 1457.0–1517.0 MHz (60 MHz continuous TDD operating bandwidth) and wideband hardware tuning capability across 1432.0–1517.0 MHz (85 MHz IBW capability; software-configurable 40/50/60/80 MHz TDD) deployed across high-density suburban and rural barangays. Siting delivery is strictly governed by the **Mandatory 3-Tier Sourcing Hierarchy**:

- **Tier 1 (Mandatory Baseline - 89.7% / 897 Sites Phase 1):** Co-location on existing operational towers (PTCI/MIDC, EdgePoint, FTAP Globe, FTAP Smart, LDIC) within 500m (urban) or 1,000m (rural) of nominal search ring centroids across the audited master tower database (13,810 registry records / 13,799 unique IDs / 13,629 coordinates).
- **Tier 2 (Secondary Target - Nominal 103 Sites):** TowerCo Build-to-Suit (B2S) where existing structures lack load capacity.

- **Tier 3 (Restricted Exception - 10.3% / 103 Sites):** Direct Operator Greenfield Monopole construction (80x 30m / 23x 40m) deployed strictly where zero TowerCos operate or all TowerCos issue formal structural denial letters.

2. DETAILED TECHNICAL SCOPE OF WORK (DEPLOYMENT & ROLLOUT SERVICES)

2.1 Site Acquisition (SAQ) & Legal Permitting Services (Standby Contingency Framework — Tier 3 Own-Build)

Applicable strictly to Tier 3 Operator Own-Build sites authorized via formal written Work Order. Evaluated in Document 06 against the 103-site Phase 1 / 515-site Program contingency basket. Scope encompasses: hunting and ranking 3 viable candidates per search ring (500m urban / 1,000m rural); spatial screening against the 13,810 audited tower database; Preliminary Candidate Evaluation Report (PCER); Site Acquisition Report (SAR) with WGS84 GPS (6 decimals), cadastral sketches, clean title verification (TCT/OCT); TowerCo Space Application (SAA) loading dossier under MLA; greenfield lease negotiations; and statutory permits: Barangay Clearance, HOA Resolution, LGU Building Permits under PD 1096, CAAP Height Clearance, DENR ECC/CNC, CFEI, FSIC, and Certificate of Occupancy.

2.2 Site Engineering & Detailed Technical Site Survey (TSSR)

Rigorous multidisciplinary engineering surveys including: 360-degree drone panoramic photo sweeps at antenna centerlines (30m to 45m AGL); compound space audit (2.0m x 2.0m pad); cable gantry routing; utility transformer kVA and DC power plant headroom audit; battery autonomy calculation (>= 4.0h @ 3.0 kW load); 3-point fall-of-potential ground resistance testing (≤ 5.0 Ohms); RF line-of-sight and clutter verification (RSRP ≥ -95 dBm); complete AutoCAD Detailed Engineering Design (DED) blueprints signed and sealed by licensed Professional Engineers; and post-construction As-Built CAD documentation.

Structural Engineering & Permitting Responsibility Matrix:

Deployment Tier	TowerCo Scope (under MLA)	Vendor Scope (Document 06 / SOW)	Permitting & PE Wet-Seal Ownership
Tier 1: Existing Tower Co-Location (897 Sites / 89.7%)	Complete 3D structural analysis under TIA-222-H (250–315 km/h site-specific basis), foundation verification, structural modification (if needed), PE wet-seal certification.	Physical site survey (TSSR), mount inspection, active equipment loading dossier (weight, EPA, brackets), joint audit with TowerCo.	TowerCo provides certified SAR with PE wet-seal under MLA. Vendor accepts mount interface.
Tier 2: Build-to-Suit (B2S) Tower (Nominal 103 Sites)	Complete tower structural design, foundation civil design, TIA-222-H certification, local building permits, PE wet-seal certification.	Provide RAN antenna load specifications, cable ladder requirements, DC power requirements.	TowerCo delivers turnkey permitted tower structure.
Tier 3: Operator Own-Build Contingency (Standby Rate Card)	N/A (Operator/Vendor execution).	**Full vendor structural engineering scope (ENG-02, ENG-06):** Complete structural wind modeling (250–315 km/h basis), foundation design, and licensed Philippine Professional Civil/Structural Engineer (PE) wet-seal.	Vendor turnkey under Standby Rate Card (ENG-02 / CALL-06).

2.3 Radio Network Planning & Design (RNPD)

Macro-level 5G RF planning utilizing 3D ray-tracing propagation modeling engines calibrated against 5m clutter and DTM data; nominal search ring generation; coverage prediction mapping; capacity dimensioning for Band n50; and Cell Design Data (CDD) parameter engineering, including cell naming, gNodeB IDs, Physical Cell ID (PCI) planning with modulo-3/modulo-30 separation,

Root Sequence Index (RSI), Neighbor Cell Lists (NCL), and eRET profiles.

2.4 Telecom Logistics, Warehousing & Material Handling

Central bonded climate-controlled warehousing in Mega Manila and regional hubs (North Luzon, South Luzon, Visayas, Mindanao); serial-number level barcode asset tagging; ERP tracking; staging site kits (1x BBU, 3x 8T8R RRUs, 3x macro antennas, optical patchcords, DC cables, SPDs, grounding accessories); pre-cutting cables; and long-haul and final-mile rough-road transport to remote barangay sites utilizing heavy-duty 4x4 trucks, motorized bancas, or manual porterage across unpaved terrain, fully insured.

2.5 Telecom Installation, Testing, Commissioning & Integration (TI)

Rigorous execution by certified riggers and telecom engineers: hoisting and mounting 3x macro high-gain antennas and 3x 8T8R RRUs on tower AMBs; setting mechanical tilts and azimuths (0°, 120°, 240°); installing eRET cables (AISG 2.0); routing armored optical jumpers and DC power cables along cable ladders; installing cold-shrink weatherproof boots; installing Type 2 DC SPDs at tower top and base; bonding 3-point grounding kits to external ground busbar; 19" rack installation of BBU; GPS antenna installation for PTP/SyncE; DC turn-up; software baseline loading; optical transceiver power verification; VSWR sweep testing (VSWR ≤ 1.35:1, Return Loss ≥ 16.5 dB); IP backhaul integration; Core (UPF/AMF) connection; OSS synchronization; alarm clearance; and commercial on-air declaration.

2.6 Single Site Verification (SSV) & Site Acceptance (ATP)

Mandatory three-stage acceptance procedure: ATP-A (Mechanical, Civil & Electrical inspection of bolt torque marks, cable dressing, weatherproofing, grounding resistance ≤ 5.0 Ohms); ATP-B (Active Telecom & Functional verification of 8T8R MU-MIMO operation, remote eRET tilt control, alarm failover simulations); and SSV (stationary throughput testing at near, mid, and cell-edge points per sector validating ≥ 450 Mbps peak DL, ≥ 50 Mbps cell-edge DL, ≥ 15 Mbps cell-edge UL, sector swap checks, handover tests, signed SSV Report, and latency verification):

- **Terrestrial Fiber Backhaul:** Target RTT **≤ 18 ms** to Core UPF; Contractual Acceptance Rejection Ceiling **≤ 30 ms** (per Appendix C KPI-RAN-07 and Bid Bulletin 01 §2.3). Sites measuring ≤ 30 ms meet provisional acceptance; sites > 30 ms require route re-engineering.
- **LEO Satellite Backhaul (Starlink / Expansion Batches):** Target RTT **≤ 60 ms**; Contractual Acceptance Rejection Ceiling **≤ 80 ms** (strictly per Bid Bulletin 01 §2.3).

2.7 Cluster Radio Network Optimization (RNO)

Multi-site cluster optimization across contiguous clusters of 10–20 sites (65 clusters Phase 1 / 325 clusters Program): pre-launch scanner and test UE drive testing; electrical downtilt (eRET) optimization; azimuth adjustments; PCI re-planning; neighbor list cleanup; handover parameter tuning; and 14-day post-launch OSS statistical validation confirming cluster Radio Availability ≥ 99.95%, Session Setup Success Rate (CSSR) ≥ 97.00% (min 96.00%), Session Drop Rate (CDDR) ≤ 1.50% (max 2.00%), and cluster average throughput ≥ 450 Mbps (harmonized w/ Bid Bulletin No. 2).

3. OPERATIONS & MAINTENANCE (O&M) / MANAGED SERVICES SPECIFICATION

The Contractor shall deliver an integrated 24/7/365 Managed Services organization to ensure continuous 99.95% network availability across all 5,000 sites nationwide:

- **24/7/365 Central NOC Surveillance:** Tier-1 real-time surveillance across RAN, Core, Power, and Transmission via EMS/OSS. Automated alarm correlation, deduplication, and trouble ticket creation within 5 minutes. Daily Operational Flash Reports at 06:00 PHT, weekly reviews, and monthly SLA dossiers.
- **Tier-2 / Tier-3 Support:** 24/7 remote diagnostics, soft/hard resets, config restoration, and transmission failover. Tier-3 OEM TAC Level 3 escalation, bug patch testing, and Root Cause Analysis (RCA) reporting within 24 hours.

- **Field First Line Maintenance (FLM):** Dedicated 24/7 mobile crews in 4x4 vehicles stationed at regional depots. Mandated MTTR: **Severity 1 (Critical) ≤ 2.0h Urban / ≤ 4.0h Rural; Severity 2 (Major) ≤ 4.0h Urban / ≤ 6.0h Rural; Severity 3 (Minor) ≤ 12.0h Urban / ≤ 24.0h Rural.** Includes active hardware swapping, passive electrical triage, and emergency fiber backhaul splicing.
- **Quarterly Preventive Maintenance (PM):** 100% completion p.a. logged via Mobile App: Module A (Telecom Equipment & Optics); Module B (DC Power Plant & LiFePO4 Battery discharge test); Module C (Grounding grid ≤ 5.0 Ohms); Module D (Bi-annual tower climbing & rigging audit); Module E (Compound civil, fencing & weed clearance).
- **Spare Parts Management (SPMS):** Regional depots maintaining an audited 2.0% active hardware float. Guaranteed Advance Hardware Replacement (AHR) Next-Business-Day (NBD) standard or 4-hour emergency delivery. Full RMA reverse logistics lifecycle tracking (TAT ≤ 30 days).
- **Continuous RNO & Interference Hunting:** Ongoing RF KPI surveillance, weekly Worst Performing Cell (WPC) triage, spectrum analyzer sweeps for external jamming/boosters, and PIM testing (≤ -155 dBc).
- **Facility, Power & Software Maintenance:** Tower anti-corrosion painting, plinth crack repair, utility Meralco/Coop coordination, 250-hour generator servicing, maintenance window patch deployments (00:00-05:00), and weekly automated backups.

4. MASTER SERVICES PRICING SCHEDULE & COMMERCIAL RATE CARD

Bidders shall complete all firm unit rates in US Dollars (USD) in Document 06 (*06_RFP_Commercial_Price_Book_and_BoQ_Template.xlsx*). Subtotals and multi-year evaluation amounts are calculated via official workbook formulas.

Item Ref	Service Title	Detailed Technical Scope & Boundaries	UoM	Ph 1 Qty	Prog Qty	Evaluation Basis	Unit Rate	Lot / TCO Mapping
DOMAIN: SITE ACQUISITION & LEGAL PERMITTING (STANDBY CONTINGENCY RATE CARD — TIER 3 OWN-BUILD 10.3% EVALUATION BASKET)								

Item Ref	Service Title	Detailed Technical Scope & Boundaries	UoM	Ph 1 Qty	Prog Qty	Evaluation Basis	Unit Rate	Lot / TCO Mapping
SAQ-01	**Nominal Search Ring Siting & Candidate Hunting**	[OPERATIONAL BASELINE: Siting and candidate hunting is executed by the TowerCo. Vendor performs on-site technical vetting (RF LOS, antenna height, structural feasibility) and issues joint endorsement; Operator issues final approval. COMMERCIAL PRICING BASIS: Quoted rate serves strictly as a Standby / Contingency Call-Off Rate Card for Operator Self-Acquisition / Own-Build]. Field identification, screening, and ranking of three (3) viable candidates (Candidate A - Primary, Candidate B - Secondary, Candidate C - Contingency) within the assigned nominal search ring (500m radius for urban/suburban; 1,000m radius for rural). Mandatory spatial screening against audited TowerCo database (13,810 audited tower locations) to prioritize Tier 1 co-location. Scope includes visual inspection of antenna mounting feasibility, ground space availability (minimum 2.0m x 2.0m pad), line-of-sight to demand settlement centroid, road access verification (all-weather vs 4x4), and issuance of the Preliminary Candidate Evaluation Report (PCER). • **Inclusions:** Desktop spatial analysis, physical on-ground hunting, 3 candidate identification, GPS logging, PCER compilation. • **Exclusions:** Commercial lease execution (handled by Operator under MLA).	Per Search Ring	103	515	1x (10.3% Own-Build Basket)	[Bidder Quote]	Lot 5 Standby Call-Off (103 Sites)

Item Ref	Service Title	Detailed Technical Scope & Boundaries	UoM	Ph 1 Qty	Prog Qty	Evaluation Basis	Unit Rate	Lot / TCO Mapping
SAQ-02	**Detailed Site Acquisition Report (SAR) & Technical Dossier**	[OPERATIONAL BASELINE: TowerCo provides primary site attributes and property profile under MLA. Vendor reviews and compiles technical feasibility annex in TSSR; Operator signs off. COMMERCIAL PRICING BASIS: Standby / Contingency Call-Off Rate Card for full standalone SAR compilation during Operator Self-Acquisition]. Comprehensive on-ground site acquisition audit and property investigation for the selected primary candidate. Compilation of standardized Site Acquisition Report (SAR) containing: official PSGC code, precision WGS84 GPS coordinates (>= 6 decimal places), ground elevation AMSL, property cadastral boundary sketch, certified land title verification (TCT/OCT/Tax Declaration), lessor profile and contact matrix, preliminary structural condition assessment, preliminary power availability (grid connection point, transformer distance, utility provider), 360-degree panoramic site photos, logistics route access classification, and SAQ risk assessment matrix. • **Inclusions:** On-ground cadastral audit, title verification, lessor interview, 360-deg photo sweep, complete SAR package. • **Exclusions:** Payment of land rental / lease deposits (direct Operator responsibility).	Per Candidate Site	103	515	1x (10.3% Own-Build Basket)	[Bidder Quote]	Lot 5 Standby Call-Off (103 Sites)

Item Ref	Service Title	Detailed Technical Scope & Boundaries	UoM	Ph 1 Qty	Prog Qty	Evaluation Basis	Unit Rate	Lot / TCO Mapping
SAQ-03	**TowerCo Space Application (SAA) & Technical Dossier Support**	[COMMITTED TIER 1 CO-LOCATION SCOPE (897 SITES)]: Preparation of technical equipment loading profile (3x 8T8R RRU, 3x Antennas, hybrid cables, 2m x 2m pad) and coordination with TowerCo engineering until SAA approval. Commercial lease executed directly by Operator under MLA. Preparation, technical compilation, submission, and engineering follow-through of the TowerCo Space Application (SAA) package for co-location under Operator's Master Lease Agreement (MLA). Compiles exact proposed loading profile: 3x 8T8R RRU, 3x macro antennas/AAUs, antenna mounting brackets (AMB), hybrid trunk cables, DC distribution requirements, and ground plinth reservation (2.0m x 2.0m). Coordinates with TowerCo technical team during structural vetting until formal issuance of SAA Approval and Notice to Proceed (NTP). • **Inclusions:** SAA documentation, loading calculation submission, liaison with TowerCo engineering, tracking SAA to NTP. • **Exclusions:** Commercial MLA negotiations and monthly tower rental fees (exclusive Operator responsibility).	Per Co-Location Site	**897**	**4,485**	89.7% Co-Location Basket (Tier 1 TowerCo)	[Bidder Quote]	Lot 5 Committed Turnkey (897 Sites Ph1 / 4,485 Prog)
SAQ-04	**Greenfield Land Lease Negotiation & Notarized Contract Support**	[CONTINGENCY / SELF-BUILD SCOPE: Applicable strictly to Operator Own-Build Greenfield sites where no TowerCo exists or TowerCo denies build. Direct lessor negotiation, title due diligence, and lease notarization]. Direct lessor negotiations for greenfield sites: boundary survey coordination, verification of registered ownership (clean title check at Registry of Deeds, inheritance/estate verification, real property tax clearance), negotiation of long-term lease terms (10-15 years) within Operator budgetary caps, preparation of lease annexes, securing lessor spousal consent and IDs, and facilitation of formal contract notarization and annotation on title. • **Inclusions:** Lessor negotiations, title due diligence, boundary verification, notary liaison, lease execution assistance. • **Exclusions:** Actual lease rentals, advance deposits, and title annotation official fees (Operator pays directly).	Per Greenfield Site	**103**	**515**	1x (10.3% Own-Build Basket)	[Bidder Quote]	Lot 5 Standby Call-Off (103 Sites)

Item Ref	Service Title	Detailed Technical Scope & Boundaries	UoM	Ph 1 Qty	Prog Qty	Evaluation Basis	Unit Rate	Lot / TCO Mapping
SAQ-05	**Local Community, Barangay & Neighborhood Clearances**	[OPERATIONAL BASELINE: TowerCo secures community/Barangay clearances for TowerCo towers. COMMERCIAL PRICING BASIS: Standby / Contingency Call-Off Rate Card for Operator Self-Acquisition / Greenfield sites]. Securing official local community clearances: presentation to Barangay Council, public hearing facilitation (if mandated), processing and issuance of official Barangay Resolution of Support, Barangay Construction Clearance, and certified Neighborhood / Homeowners Association (HOA) Resolution / Certificate of No Objection from contiguous property owners within a 50-meter radius. • **Inclusions:** Barangay consultations, neighbor signature campaigns, clearance filing and follow-through, official receipts. • **Exclusions:** Official LGU statutory fees (reimbursable at actual cost against official receipts).	Per Site	**103**	**515**	1x (10.3% Own-Build Basket)	[Bidder Quote]	Lot 5 Standby Call-Off (103 Sites)
SAQ-06	**Local Government Unit (LGU) Building & Construction Permits**	[OPERATIONAL BASELINE: TowerCo secures LGU Building Permits under PD 1096 for TowerCo towers. COMMERCIAL PRICING BASIS: Standby / Contingency Call-Off Rate Card for Operator Self-Build Greenfield Monopoles]. Complete processing and procurement of municipal/city LGU Building Permits in accordance with the National Building Code of the Philippines (PD 1096). Scope includes securing Locational Clearance, Zoning Compliance, Architectural Permit, Civil/Structural Permit, Electrical Permit, Sanitary/Plumbing Permit, and Mechanical Permit, backed by complete sets of plans signed and sealed by licensed Professional Engineers (Civil, Structural, Electrical, PEE). • **Inclusions:** Permit application filing, engineering plan liaison, follow-through across city/municipal engineering offices. • **Exclusions:** Official municipal permit assessment fees (reimbursable at face value against official receipts).	Per Greenfield Site / Co-loc where required	**103**	**515**	1x (10.3% Own-Build Basket)	[Bidder Quote]	Lot 5 Standby Call-Off (103 Sites)

Item Ref	Service Title	Detailed Technical Scope & Boundaries	UoM	Ph 1 Qty	Prog Qty	Evaluation Basis	Unit Rate	Lot / TCO Mapping
SAQ-07	**Statutory & Environmental Clearances (CAAP, DENR, DPWH)**	[OPERATIONAL BASELINE: TowerCo secures CAAP Height Clearance and DENR CNC/ECC for TowerCo assets. COMMERCIAL PRICING BASIS: Standby / Contingency Call-Off Rate Card for Operator Self-Build Greenfield Monopoles]. Procurement of mandatory national statutory clearances: Civil Aviation Authority of the Philippines (CAAP) Height Clearance Permit / Obstruction Evaluation; Department of Environment and Natural Resources (DENR) Certificate of Non-Coverage (CNC) or Environmental Compliance Certificate (ECC); and Department of Public Works and Highways (DPWH) Road Right-of-Way (ROW) clearance where site access fronts national highways. • **Inclusions:** Technical submittals, coordinate and elevation certification, CAAP/DENR filing and technical defense. • **Exclusions:** Official statutory assessment fees (reimbursable at actual cost).	Per Site	103	515	1x (10.3% Own-Build Basket)	[Bidder Quote]	Lot 5 Standby Call-Off (103 Sites)
SAQ-08	**Utility Interconnection, CFEI, FSIC & Occupancy Permitting**	[OPERATIONAL BASELINE: TowerCo manages utility service entrance and meter installation on shared compound. COMMERCIAL PRICING BASIS: Standby / Contingency Call-Off Rate Card for Operator Self-Build sites]. Securing post-construction occupancy and power clearances: processing Certificate of Final Electrical Inspection (CFEI), Bureau of Fire Protection (BFP) Fire Safety Inspection Certificate (FSIC), commercial power utility (Meralco / Regional Electric Cooperative) service application, transformer tapping coordination, meter base energization, and municipal LGU Certificate of Final Occupancy / Permit to Operate. • **Inclusions:** BFP inspection facilitation, CFEI filing, utility coordination, meter installation liaison, occupancy release. • **Exclusions:** Utility transformer installation capital contribution and security deposits (paid by Operator).	Per Site	103	515	1x (10.3% Own-Build Basket)	[Bidder Quote]	Lot 5 Standby Call-Off (103 Sites)
DOMAIN: SITE ENGINEERING & DETAILED SITE SURVEY								

Item Ref	Service Title	Detailed Technical Scope & Boundaries	UoM	Ph 1 Qty	Prog Qty	Evaluation Basis	Unit Rate	Lot / TCO Mapping
ENG-01	**Technical Site Survey (TSS) & 360° Drone Panoramic Audit**	Rigorous multi-disciplinary physical site survey executed by licensed telecom and civil engineers: 360° high-resolution panoramic photographic sweep at target antenna centerline (using calibrated drone or mast camera); precision dual-frequency GPS coordinate recording; physical measurement of compound dimensions, tower base, shelter/cabinet footprint; cable gantry routing audit; transport access route assessment (bridge load limits, tight turning radii); and compilation of raw TSS data. • **Inclusions:** Multi-disciplinary field team, calibrated drone flight, 360-deg panoramic photos, spatial sketches, access log. • **Exclusions:** Tower climbing above 60m (requires separate high-mast specialized rigging permit if applicable).	Per Site	1,000	5,000	1x (Turnkey Rollout)	[Bidder Quote]	Lot 5 Turnkey Rollout Baseline
ENG-02	**Site Structural Audit, Antenna Loading Verification & Own-Build Structural Engineering**	Engineering structural evaluation of the host structure: [OPERATIONAL BASELINE — TIER 1 CO-LOCATION & TIER 2 B2S: Under MLA, TowerCo conducts 3D structural analysis, TIA-222-H wind modeling, foundation verification, and secures the licensed PE wet-seal certification per Appendix D §1.3 and Bid Bulletin 01 Huawei Item 13. Vendor scope is strictly limited to physical audit of existing mounts, active equipment loading profile calculation (weight, EPA, mounting dimensions for 3x 8T8R RRU, 3x macro antennas/AAUs, AMB brackets, hybrid trunk cable), and mounting interface verification. STANDBY BASELINE — TIER 3 OPERATOR OWN-BUILD: Vendor provides full turnkey 3D finite-element structural wind loading simulation based on TIA-222-H / NSCP 2015 site-specific basic wind speed (250 km/h to 315 km/h, Category 5 typhoon standard), foundation capacity verification, and formal Structural Analysis Report wet-sealed by a licensed Philippine Professional Civil/Structural Engineer]. • **Inclusions:** Physical mount audit, active equipment EPA/weight loading dossier, interface sign-off; (for Tier 3 Own-Build: full 250-315 km/h TIA-222-H wind modeling, foundation check, licensed PE wet-seal). • **Exclusions:** Tower reinforcement civil execution (governed by separate structural modification call-out rate card CALL-06); TowerCo structural analysis on leased towers (TowerCo MLA scope).	Per Site	1,000	5,000	1x (Turnkey Rollout)	[Bidder Quote]	Lot 5 Turnkey Rollout Baseline

Item Ref	Service Title	Detailed Technical Scope & Boundaries	UoM	Ph 1 Qty	Prog Qty	Evaluation Basis	Unit Rate	Lot / TCO Mapping
ENG-03	**Electrical, DC Power & Earth Grounding Engineering Assessment**	Electrical engineering audit of commercial AC utility power and DC power plant: inspection of transformer capacity (kVA), service entrance voltage, phase balance, AC main breaker rating; assessment of existing DC power plant (rectifier total capacity, SMPS controller, available DC branch breakers >= 63A for new radio load); calculation of LiFePO4 battery autonomy (minimum 4.0h @ 3.0 kW load, <= 80% DoD); and 3-point fall-of-potential earth resistance measurement of the grounding grid, mandating <= 5.0 Ohms pursuant to Philippine Electrical Code (PEC) and TelTech engineering specifications. • **Inclusions:** Commercial power audit, DC rectifier capacity calculation, battery autonomy simulation, 3-point earth test. • **Exclusions:** Major utility transformer upgrades (handled under utility capex contribution).	Per Site	1,000	5,000	1x (Turnkey Rollout)	[Bidder Quote]	Lot 5 Turnkey Rollout Baseline
ENG-04	**RF Engineering Siting, Line-of-Sight (LOS) & Clutter Assessment**	On-site RF validation of nominal design coverage over target barangay demand centroid: physical line-of-sight (LOS) inspection at proposed antenna centerlines (30m, 35m, 40m AGL); clutter height and foliage density classification; identification of close-in obstructions within 200m of sector azimuths (0°, 120°, 240°); calculation of first Fresnel zone clearance; and field verification that simulated coverage will achieve target cell-edge RSRP >= -95 dBm and SINR >= 6 dB. • **Inclusions:** On-ground visual LOS audit, clutter verification, azimuth clearance check, RF siting validation report. • **Exclusions:** Continuous cluster drive test (governed under Lot 5 Scope RNO-01).	Per Site	1,000	5,000	1x (Turnkey Rollout)	[Bidder Quote]	Lot 5 Turnkey Rollout Baseline

Item Ref	Service Title	Detailed Technical Scope & Boundaries	UoM	Ph 1 Qty	Prog Qty	Evaluation Basis	Unit Rate	Lot / TCO Mapping
ENG-05	**Detailed Engineering Design (DED) & Technical Site Survey Report (TSSR)**	Consolidation of multidisciplinary survey data into a complete Detailed Engineering Design (DED) blueprint package and Technical Site Survey Report (TSSR): Architectural Site Layout Plan; Tower Elevation & Antenna Orientation Plan; Antenna Mounting Bracket (AMB) Fabrication Details; Cable Ladder, Gantry & Cable Tray Routing Plan; Electrical Single Line Diagram (SLD); Earth Grounding Ring & Lightning Protection Layout; Site Bill of Quantities (BoQ); and formal endorsement by licensed Professional Engineers (PE-Civil/Structural, PE-Electrical, PECE-Electronics) conforming to NBCP and PEC. • **Inclusions:** AutoCAD DED drawings, multi-PE signed & sealed plans, BoQ compilation, full TSSR dossier submission. • **Exclusions:** Structural redesigns necessitated by Operator-mandated changes to nominal search ring after survey approval.	Per Site	1,000	5,000	1x (Turnkey Rollout)	[Bidder Quote]	Lot 5 Turnkey Rollout Baseline
ENG-06	**As-Built Engineering Documentation & Final Completion Dossier**	Comprehensive post-construction field engineering audit capturing all redline installation adjustments: updating AutoCAD drawings to reflect exact as-built antenna heights, azimuths, mechanical tilts, optical cable routes, rack layouts, breaker allocations, and grounding busbar termination points; compiling comprehensive photographic completion book (minimum 50 high-res timestamped photos per site); and securing final Professional Engineer endorsement. • **Inclusions:** Field redline audit, final CAD as-built drawings (DWG & PDF), photographic completion book, PE endorsement. • **Exclusions:** Routine operational documentation updates post-commercial acceptance.	Per Site	1,000	5,000	1x (Turnkey Rollout)	[Bidder Quote]	Lot 5 Turnkey Rollout Baseline
DOMAIN: RADIO NETWORK PLANNING & DESIGN								

Item Ref	Service Title	Detailed Technical Scope & Boundaries	UoM	Ph 1 Qty	Prog Qty	Evaluation Basis	Unit Rate	Lot / TCO Mapping
RNPD–01	**Macro RF Nominal Planning & 3D Propagation Modeling**	Macro-level 5G RF network planning using advanced 3D ray-tracing propagation modeling engines, calibrated against high-resolution 5-meter clutter models and Digital Terrain Models (DTM). Sizing and generation of nominal candidate search rings across high-density suburban and rural barangay centroids; multi-carrier capacity dimensioning for Band n50 based on the authorized 60 MHz continuous TDD baseline (1457.0–1517.0 MHz) nationwide for contractual evaluation, with future software-expandable sensitivity modeling up to 80 MHz in provincial clusters (pursuant to Bulletin No. 2 §3.1); coverage prediction mapping (RSRP, SINR, Best Server, Downlink/Uplink Throughput); and inter-site distance (ISD) optimization. • **Inclusions:** 3D ray-tracing simulation, clutter model calibration, nominal site planning, search ring generation. • **Exclusions:** Procurement of specialized external digital terrain data licenses (provided by Operator).	Per Site	1,000	5,000	1x (Turnkey Rollout)	[Bidder Quote]	Lot 5 Turnkey Rollout Baseline
RNPD–02	**Cell Design Data (CDD) Engineering & Parameter Allocation**	Engineering, generation, and validation of detailed site-specific Cell Design Data (CDD) sheets: cell naming taxonomy, gNodeB IDs, Physical Cell ID (PCI) planning adhering to modulo-3 and modulo-30 collision rules, Root Sequence Index (RSI), Tracking Area Codes (TAC), Neighbor Cell Lists (NCL / ANR), PRACH configuration, absolute radio-frequency channel numbers (ARFCN), per-carrier transmit power, and antenna configuration parameters (azimuth, mechanical tilt, eRET ranges). • **Inclusions:** CDD parameter engineering, PCI planning, neighbor list generation, configuration script generation. • **Exclusions:** Core network routing configuration (handled under Lot 3).	Per Site	1,000	5,000	1x (Turnkey Rollout)	[Bidder Quote]	Lot 5 Turnkey Rollout Baseline
DOMAIN: TELECOM LOGISTICS & MATERIAL HANDLING								

Item Ref	Service Title	Detailed Technical Scope & Boundaries	UoM	Ph 1 Qty	Prog Qty	Evaluation Basis	Unit Rate	Lot / TCO Mapping
LOG-01	**Central & Regional Warehousing, Buffer Staging & Serialized Asset Tagging**	Receipt of OEM shipments at Port of Manila / Batangas; bonded climate-controlled warehousing in central Mega Manila and regional hubs (North Luzon, South Luzon, Visayas, Mindanao); unboxing, physical inspection, and QA damage checking; barcode asset tagging at serial-number level; updating Operator ERP/WMS inventory records; and maintaining segregated buffer stock. • **Inclusions:** Warehouse facility, security, ERP tracking, barcode tagging, climate control, buffer stock management. • **Exclusions:** Port customs duties and VAT taxes (direct Operator responsibility).	Per Site	1,000	5,000	1x (Turnkey Rollout)	[Bidder Quote]	Lot 5 Turnkey Rollout Baseline
LOG-02	**Site BOM Kitting, Staging & Pre-Assembly Services**	Assembly and kitting of site-specific deployment packages: picking 1x BBU, 3x 8T8R RRUs, 3x macro antennas/AAUs, optical patchcords, DC power cables, Type 2 SPDs, grounding kits, and hardware accessories; pre-cutting cable lengths based on approved TSSR; labeling; staging in protective crates; and performing bench test pre-checks and firmware pre-flashing. • **Inclusions:** Picking, staging, pre-cutting cables, labeling, firmware pre-flashing, packaging into site kits. • **Exclusions:** Supply of missing OEM radio equipment (OEM responsibility).	Per Site	1,000	5,000	1x (Turnkey Rollout)	[Bidder Quote]	Lot 5 Turnkey Rollout Baseline
LOG-03	**Regional Transport & Final-Mile Rough-Road Site Hauling**	Transportation from central warehouse to regional hubs via heavy freight, followed by dedicated final-mile transport to remote barangay site locations utilizing heavy-duty 4x4 trucks, motorized bancas/barges (for island sites), or manual porterage across unpaved rough-road terrain; comprehensive in-transit cargo insurance; and on-site offloading. • **Inclusions:** Long-haul freight, final-mile 4x4 hauling, manual porterage, transit insurance, on-site offloading. • **Exclusions:** Helicopter airlifts (applicable only under extraordinary force majeure call-out).	Per Site	1,000	5,000	1x (Turnkey Rollout)	[Bidder Quote]	Lot 5 Turnkey Rollout Baseline
DOMAIN: TELECOM INSTALLATION, COMMISSIONING & INTEGRATION								

Item Ref	Service Title	Detailed Technical Scope & Boundaries	UoM	Ph 1 Qty	Prog Qty	Evaluation Basis	Unit Rate	Lot / TCO Mapping
TI-01	**Tower Rigging & Mechanical Antenna / RRU Installation**	Mechanical mounting by certified riggers: hoisting and installing 3x macro high-gain antennas and 3x 8T8R RRUs on tower antenna mounting brackets (AMB) at specified AGL centerlines (30m to 45m); setting precise mechanical downtilt and sector azimuths (0°, 120°, 240°) using calibrated digital inclinometers and compasses; and installing AISG 2.0 eRET control cables. • **Inclusions:** Tower rigging crews, safety equipment, hoisting gear, AMB mounting, antenna and RRU installation. • **Exclusions:** Supply of steel antenna mounting brackets (AMB) (governed under Lot 1 hardware).	Per Site	1,000	5,000	1x (Turnkey Rollout)	[Bidder Quote]	Lot 5 Turnkey Rollout Baseline
TI-02	**RF, Optical Fiber & DC Power Cabling Installation**	Routing, fastening, and dressing armored multi-core optical fiber jumpers and DC power cables along tower cable ladders and horizontal gantries; installing cold-shrink weatherproof boots on all 7/16 DIN or 4.3-10 RF connectors; installing Type 2 DC Surge Protection Devices (SPD) at tower top and base; and bonding 3-point grounding kits to the external ground busbar. • **Inclusions:** Cable routing, cleating, hydraulic crimping, weatherproofing, SPD installation, grounding bond. • **Exclusions:** Internal civil cabling of third-party shelters.	Per Site	1,000	5,000	1x (Turnkey Rollout)	[Bidder Quote]	Lot 5 Turnkey Rollout Baseline
TI-03	**Baseband Unit (BBU) Installation & Cabinet Wiring**	19" rack installation of 5G Baseband Unit (BBU) in outdoor equipment cabinet or shelter; dual -48V DC power wiring to dedicated distribution breakers; routing backhaul transmission optical patchcords to fiber demarcation router; and installing GNSS/GPS antenna and surge arrester with unobstructed sky view for PTP/SyncE clock synchronization. • **Inclusions:** Rack mounting, DC wiring, optical patchcord routing, GPS antenna installation, labeling. • **Exclusions:** Supply of DC rectifier plant (governed under Lot 2).	Per Site	1,000	5,000	1x (Turnkey Rollout)	[Bidder Quote]	Lot 5 Turnkey Rollout Baseline

Item Ref	Service Title	Detailed Technical Scope & Boundaries	UoM	Ph 1 Qty	Prog Qty	Evaluation Basis	Unit Rate	Lot / TCO Mapping
TI-04	**Telecom Commissioning, Sweep Testing & Power-Up**	DC power turn-up, voltage and polarity verification; firmware loading and diagnostic testing; optical transceiver Rx/Tx power level verification; and calibrated RF sweep testing measuring Return Loss (>= 16.5 dB) and VSWR (<= 1.35:1) across all antenna ports and jumpers using a calibrated SiteMaster analyzer. • **Inclusions:** Commissioning turn-up, firmware loading, optical power check, VSWR sweep testing, test trace logs. • **Exclusions:** Rectification of OEM internal hardware manufacturing defects (OEM warranty scope).	Per Site	1,000	5,000	1x (Turnkey Rollout)	[Bidder Quote]	Lot 5 Turnkey Rollout Baseline
TI-05	**gNodeB Network Integration & OSS Commercial On-Air**	Transmission IP integration (VLAN, BGP, IPsec); establishing S1-U / N2/N3 control and user plane interfaces to 5G Core (AMF/UPF); synchronization with TelTech EMS/OSS; activating 3 sectors on Band n50 on the authorized 60 MHz continuous TDD carrier (1457.0–1517.0 MHz, with hardware wideband tuning across 1432.0–1517.0 MHz and software-configurable support up to 80 MHz TDD pursuant to Bulletin No. 2 §3.1); clearing all active alarms; and achieving stable commercial on-air operational state. • **Inclusions:** IP configuration, Core integration, OSS synchronization, carrier activation, alarm clearance. • **Exclusions:** Core network UPF dimensioning (governed under Lot 3).	Per Site	1,000	5,000	1x (Turnkey Rollout)	[Bidder Quote]	Lot 5 Turnkey Rollout Baseline
DOMAIN: SINGLE SITE VERIFICATION & ACCEPTANCE								
ACC-01	**Mechanical, Civil & Electrical Site Acceptance (ATP-A)**	Joint physical on-site inspection: verifying bolt torque marks on tower mounts, cable ladder fastening, cable tray grounding, weatherproofing seal integrity, electrical breaker labeling, 3-point earth ground resistance audit (confirming <= 5.0 Ohms), compound housekeeping, and punchlist compilation and clearance. • **Inclusions:** Joint physical audit, torque inspection, grounding verification, punchlist clearance. • **Exclusions:** Radio performance verification (covered under ATP-B and SSV).	Per Site	1,000	5,000	1x (Turnkey Rollout)	[Bidder Quote]	Lot 5 Turnkey Rollout Baseline

Item Ref	Service Title	Detailed Technical Scope & Boundaries	UoM	Ph 1 Qty	Prog Qty	Evaluation Basis	Unit Rate	Lot / TCO Mapping
ACC-02	**Active Telecom & Functional Acceptance (ATP-B)**	Remote and local active functional testing: verifying 8T8R MU-MIMO operation, electronic downtilt (eRET) remote control functionality via OSS, power fail / mains failure alarm simulation, optical fiber failover test, and carrier shutdown/restart diagnostic tests to ensure robust autonomic operation. • **Inclusions:** Remote EMS diagnostic testing, alarm simulation, eRET control validation, MIMO verification. • **Exclusions:** Physical climbing inspection (covered under ATP-A).	Per Site	1,000	5,000	1x (Turnkey Rollout)	[Bidder Quote]	Lot 5 Turnkey Rollout Baseline
ACC-03	**Single Site Verification (SSV) & RF Call Testing**	On-ground field testing using 5G test UE and scanner: stationary throughput tests at near, mid, and cell-edge points per sector (validating >= 450 Mbps peak DL, >= 50 Mbps cell-edge DL, >= 15 Mbps cell-edge UL); sector swap validation; intra-site handover execution; latency verification across backhaul media (Terrestrial Fiber: Target RTT <= 18 ms to Core UPF, with Contractual Acceptance Rejection Ceiling <= 30 ms per Appendix C KPI-RAN-07 and Bid Bulletin 01 §2.3; LEO Satellite Starlink / Expansion: Target RTT <= 60 ms, Contractual Acceptance Rejection Ceiling <= 80 ms per Bid Bulletin 01 §2.3); packet loss measurement (< 0.1%); and compilation and formal sign-off of the comprehensive SSV Report. • **Inclusions:** Stationary throughput tests, sector swap check, handover test, latency compliance test (<= 30 ms fiber ceiling / <= 80 ms satellite ceiling), SSV report compilation. • **Exclusions:** Multi-site cluster drive testing (covered under Category 7).	Per Site	1,000	5,000	1x (Turnkey Rollout)	[Bidder Quote]	Lot 5 Turnkey Rollout Baseline
DOMAIN: CLUSTER RADIO NETWORK OPTIMIZATION								
RNO-01	**Pre-Launch Cluster Drive Testing & RF Baseline Audit**	Continuous scanner and test UE drive testing across contiguous clusters of 10–20 sites; capturing RSRP, SINR, CQI, throughput, and coverage overlap; and identifying pilot pollution zones, coverage holes, and excessive inter-cell interference. • **Inclusions:** Drive test vehicle, scanner, test UEs, GPS logging, drive test raw log processing, baseline report. • **Exclusions:** Physical antenna re-orientation (governed under RNO-02).	Per Site Share	1,000	5,000	1x (Turnkey Rollout)	[Bidder Quote]	Lot 5 Turnkey Rollout Baseline

Item Ref	Service Title	Detailed Technical Scope & Boundaries	UoM	Ph 1 Qty	Prog Qty	Evaluation Basis	Unit Rate	Lot / TCO Mapping
RNO-02	**Cluster Physical & Soft Parameter Optimization**	Iterative optimization across cluster: electronic remote electrical tilt (eRET) adjustments to control cell-edge interference and shape coverage; azimuth and mechanical tilt adjustments where indicated; PCI re-planning to resolve mod-3/mod-30 clashes; neighbor list (NCL) cleanup; handover threshold tuning; and uplink power control adjustments. • **Inclusions:** eRET tuning, PCI re-planning, neighbor tuning, handover optimization, parameter upload script. • **Exclusions:** Major tower antenna re-heightening (governed by call-out rate card).	Per Site Share	1,000	5,000	1x (Turnkey Rollout)	[Bidder Quote]	Lot 5 Turnkey Rollout Baseline
RNO-03	**Cluster Acceptance KPI Validation & Final Sign-Off**	Post-tuning verification drive test and OSS 14-day statistical performance audit: validating cluster Radio Availability >= 99.95%, Session Setup Success Rate (CSSR) >= 97.00% (min 96.00%), Session Drop Rate (CDDR) <= 1.50% (max 2.00%), cluster average DL throughput >= 450 Mbps, cell-edge DL >= 50 Mbps; and submission of formal Cluster Acceptance Dossier (harmonized w/ Bid Bulletin No. 2). • **Inclusions:** Verification drive test, 14-day OSS KPI statistical audit, benchmark reporting, formal sign-off. • **Exclusions:** Ongoing post-commercial operations optimization (covered under Lot 6 Managed Services).	Per Cluster (65 Clusters Phase 1 / 325 Clusters Program)	65	325	1x (Turnkey Rollout)	[Bidder Quote]	Lot 5 Turnkey Rollout Baseline
DOMAIN: OPERATIONS & MAINTENANCE (O&M) / MANAGED SERVICES (24/7 MANAGED O&M — ANNUALIZED 12-MONTH & 3-YEAR CONTRACTUAL BASELINE)								
MS-NOC-01	**24/7/365 Central NOC Tier-1 Active Surveillance & Alarm Monitoring**	Continuous 24/7/365 active monitoring of all active and passive network elements across RAN, Core, Power, and Transmission; automated alarm correlation and noise suppression; ticket logging within 5 minutes of alarm onset; and issuance of Daily Operational Flash Report at 06:00 PHT. • **Inclusions:** 24/7 Tier-1 NOC staffing, automated correlation, trouble ticket creation, daily flash reporting. • **Exclusions:** Third-party commercial power utility outages outside contractor control (tracked as excusable SLA).	Per Site / Month	1,000	5,000	12 Months / Year	[Bidder Quote]	Lot 6 Managed Services (Retainer)

Item Ref	Service Title	Detailed Technical Scope & Boundaries	UoM	Ph 1 Qty	Prog Qty	Evaluation Basis	Unit Rate	Lot / TCO Mapping
MS-NOC-02	**Incident Management, Triage & Automated Trouble Ticketing**	End-to-end incident management: automated ticketing via OSS/ITSM; severity classification (Sev 1 Critical to Sev 4 Warning); technician dispatch initiation within 15 minutes for Sev 1 incidents; bridge coordination; and stakeholder notification. • **Inclusions:** ITSM platform, ticket lifecycle tracking, escalation dispatch, MTTR SLA auditing. • **Exclusions:** Operator customer service call center operations (BSS scope).	Per Site / Month	1,000	5,000	12 Months / Year	[Bidder Quote]	Lot 6 Managed Services (Retainer)
MS-NOC-03	**Tier-2 Remote Technical Diagnostics, Fault Recovery & Config Management**	24/7 remote engineering support: soft and hard resets, remote diagnostic sweeps, card re-initialization, backup configuration restoration, transmission routing failover, and software rollback to clear network faults without field truck rolls. • **Inclusions:** Tier-2 remote engineers, remote script execution, transmission path failover, config restoration. • **Exclusions:** Physical on-site hardware replacement (covered under FLM).	Per Site / Month	1,000	5,000	12 Months / Year	[Bidder Quote]	Lot 6 Managed Services (Retainer)
MS-NOC-04	**Tier-3 Engineering Escalation & OEM TAC Management**	Advanced engineering escalation for complex system bugs, protocol crashes, and persistent hardware/software anomalies: case logging with OEM TAC (Level 3), trace log collection, bug patch testing, software patch deployment, and Root Cause Analysis (RCA) reporting within 24 hours of major incident restoration. • **Inclusions:** OEM TAC case management, trace capture, patch verification, RCA report compilation. • **Exclusions:** OEM base software annual license royalties (governed under Software AMC).	Per Site / Month	1,000	5,000	12 Months / Year	[Bidder Quote]	Lot 6 Managed Services (Retainer)

Item Ref	Service Title	Detailed Technical Scope & Boundaries	UoM	Ph 1 Qty	Prog Qty	Evaluation Basis	Unit Rate	Lot / TCO Mapping
MS-FLM-01	**24/7 Field First Line Maintenance (FLM) Emergency Response**	Provision of dedicated, nationwide 24/7 mobile field maintenance crews equipped with 4x4 vehicles, complete safety gear, calibrated test instruments, and emergency restoration tools. Strict compliance with mandated MTTR SLAs: Severity 1 (Critical - Site Outage): Response <= 15 mins, MTTR <= 2.0h (Urban) / <= 4.0h (Rural); Severity 2 (Major - Sector Down / Power Degraded): Response <= 30 mins, MTTR <= 4.0h (Urban) / <= 6.0h (Rural); Severity 3 (Minor - Non-Service Affecting): Response <= 1h, MTTR <= 12.0h (Urban) / <= 24.0h (Rural); Severity 4 (Warning / Planned): Response <= 4h, MTTR <= 48.0h (Urban) / <= 72.0h (Rural). • **Inclusions:** Dedicated 24/7 crews, 4x4 maintenance vehicles, tools, fuel, emergency call-outs. • **Exclusions:** Force majeure acts of God (typhoon damage recovery governed under Call-Out Rate Card).	Per Site / Month	1,000	5,000	12 Months / Year	[Bidder Quote]	Lot 6 Managed Services (Retainer)
MS-FLM-02	**Active Telecom Hardware Replacement & Field Swapping**	On-site replacement of damaged or defective active telecom hardware: RRUs, BBUs, baseband processing cards, optical SFP transceivers, RF jumpers, and GPS antennas, utilizing pre-staged regional spare parts float to achieve fast-track MTTR restoration. • **Inclusions:** Technician labor, de-installation, replacement installation, power-up test, return voucher. • **Exclusions:** Cost of out-of-warranty hardware destruction caused by third-party vandalism.	Included in Base Retainer ($0.00)	1,000	5,000	Included in Retainer ($0.00)	*$0.00 (Included)*	Lot 6 Retainer Included
MS-FLM-03	**Passive Power & Infrastructure Emergency Restoration**	Rapid restoration of passive site elements: resetting/replacing tripped DC distribution breakers, rectifying rectifier module faults, clearing battery management system (BMS) lockouts, replacing blown AC/DC surge protection devices (SPDs), and diagnosing automatic transfer switch (ATS) failures. • **Inclusions:** Electrical diagnosis, breaker reset/replacement, SPD swapping, BMS diagnostic clearing. • **Exclusions:** Replacement of entire battery bank destroyed by external flood (insurance scope).	Included in Base Retainer ($0.00)	1,000	5,000	Included in Retainer ($0.00)	*$0.00 (Included)*	Lot 6 Retainer Included

Item Ref	Service Title	Detailed Technical Scope & Boundaries	UoM	Ph 1 Qty	Prog Qty	Evaluation Basis	Unit Rate	Lot / TCO Mapping
MS-FLM-04	**Emergency Fiber Backhaul Splicing & Optical Link Restoration**	Emergency mobilization for fiber transmission outages: OTDR optical fault localization, optical cable drop replacement, mechanical or fusion splicing, and optical splice enclosure re-sealing to restore backhaul transport within mandated MTTR. • **Inclusions:** OTDR fault tracing, fusion splicing labor, splice enclosure resealing, link test. • **Exclusions:** Major fiber spans exceeding 500 meters (governed by Call-Out Rate Card).	Included in Base Retainer ($0.00)	1,000	5,000	Included in Retainer ($0.00)	*$0.00 (Included)*	Lot 6 Retainer Included
MS-PM-01	**Quarterly Telecom Equipment Preventive Maintenance (Module A)**	Execution of 100% quarterly routine telecom PM: visual inspection of BBU/RRU chassis, optical fiber connector cleaning and inspection, RF jumper weatherproofing boot inspection, cable clamp torque verification, optical SFP power level verification, and dust filter cleaning. • **Inclusions:** Quarterly on-site inspection, connector cleaning, weatherproofing audit, SFP level audit, PM report. • **Exclusions:** Major equipment overhaul.	Per Site / Quarter	1,000	5,000	4 Quarters / Year	[Bidder Quote]	Lot 6 Managed Services (PM)
MS-PM-02	**Quarterly DC Power & LiFePO4 Battery PM (Module B)**	Comprehensive quarterly electrical PM: rectifier module load sharing verification, DC output float/equalize voltage calibration, terminal screw torque verification, LiFePO4 battery BMS diagnostic sweep, cell voltage and internal resistance measurement, and 1-hour controlled battery discharge autonomy test under load. • **Inclusions:** Rectifier calibration, BMS diagnostic sweep, internal resistance test, 1-hour load test. • **Exclusions:** Battery module replacement (handled under RMA / Spares float).	Per Site / Quarter	1,000	5,000	4 Quarters / Year	[Bidder Quote]	Lot 6 Managed Services (PM)
MS-PM-03	**Quarterly Earth Grounding & Lightning Protection Audit (Module C)**	Precision ground resistance audit using 3-point fall-of-potential test instrument (confirming earth ground resistance <= 5.0 Ohms); Cadweld connection inspection; copper busbar bolt torque check; corrosion audit; and inspection of air terminals, down-conductors, and surge arresters. • **Inclusions:** 3-point fall-of-potential test, Cadweld check, busbar torque check, lightning arrester audit. • **Exclusions:** Civil construction of new ground ring (if soil conditions deteriorate; governed by civil rate card).	Per Site / Quarter	1,000	5,000	4 Quarters / Year	[Bidder Quote]	Lot 6 Managed Services (PM)

Item Ref	Service Title	Detailed Technical Scope & Boundaries	UoM	Ph 1 Qty	Prog Qty	Evaluation Basis	Unit Rate	Lot / TCO Mapping
MS-PM-04	**Bi-Annual Tower Structure, Rigging & Aviation Light Audit (Module D)**	Specialized climbing inspection conducted twice per year: physical inspection of tower steel members, monopole climbing ladder, fall arrest cable, AMB mounting bracket bolt torque verification, antenna azimuth/tilt physical check, cable ladder clamping check, and inspection of CAAP-compliant aviation obstruction light and photocell controller. • **Inclusions:** Certified climbing riggers, structural member inspection, bolt torque audit, CAAP light audit. • **Exclusions:** Tower painting / rust rehabilitation (governed under civil facility maintenance).	Per Site / Semi-Annual	1,000	5,000	2 Semi-Annuals / Year	[Bidder Quote]	Lot 6 Managed Services (Tower PM)
MS-PM-05	**Quarterly Site Compound, Civil & Environmental Upkeep (Module E)**	Routine civil compound maintenance: inspection and repair of chain-link perimeter fencing and barbed wire; greasing and servicing gate locking mechanisms; weed cutting, vegetation clearance, and chemical weed-killing within a 3-meter perimeter of compound; drainage channel clearing; and pest/rodent proofing of equipment cabinet cable entries. • **Inclusions:** Compound weeding, perimeter clearing, fence repair, lock lubrication, cable entry pest sealing. • **Exclusions:** Major civil perimeter wall re-construction post-landslide.	Per Site / Quarter	1,000	5,000	4 Quarters / Year	[Bidder Quote]	Lot 6 Managed Services (PM)
MS-SPM-01	**Regional Spare Parts Buffer Float Management (SPMS)**	Operation of climate-controlled, secure regional spare parts depots (Mega Manila, North Luzon, South Luzon, Visayas, Mindanao); managing and maintaining an agreed 2.0% active hardware spares float (BBUs, RRUs, SFPs, rectifiers, batteries, cables); tracking serial numbers via WMS; and conducting monthly physical cycle counts. • **Inclusions:** Regional storage depots, inventory management, serialized tracking, monthly cycle counts. • **Exclusions:** Procurement of additional replacement spares beyond contractual float (Operator capex).	Per Site / Month	1,000	5,000	12 Months / Year	[Bidder Quote]	Lot 6 Managed Services (Retainer)
MS-SPM-02	**Advance Hardware Replacement (AHR) & Expedited Field Delivery**	Rapid logistics dispatch delivering replacement hardware to field maintenance crews: Next-Business-Day (NBD) standard delivery nationwide, or expedited 4-hour emergency delivery for major macro hubs and critical transmission nodes. • **Inclusions:** Expedited dispatch, emergency courier, chain of custody documentation, field delivery receipt. • **Exclusions:** Standard scheduled routine replenishment.	Included in Base Retainer ($0.00)	1,000	5,000	Included in Retainer ($0.00)	*$0.00 (Included)*	Lot 6 Retainer Included

Item Ref	Service Title	Detailed Technical Scope & Boundaries	UoM	Ph 1 Qty	Prog Qty	Evaluation Basis	Unit Rate	Lot / TCO Mapping
MS-SPM-03	**Return Merchandise Authorization (RMA) & OEM Factory Repair Logistics**	Managing the reverse logistics lifecycle for defective hardware: retrieving faulty units from field crews, conducting triage, generating RMA requests, managing packaging and export/import customs for OEM repair/replacement, tracking turnaround time (TAT <= 30 calendar days), and returning repaired units to the regional buffer float. • **Inclusions:** Defective unit triage, RMA processing, shipping to OEM repair facility, tracking, customs clearance. • **Exclusions:** Out-of-warranty OEM repair billing charges (passed through at actual OEM repair cost).	Included in Base Retainer ($0.00)	1,000	5,000	Included in Retainer ($0.00)	*$0.00 (Included)*	Lot 6 Retainer Included
MS-RNO-01	**Continuous Radio Network Optimization & Weekly KPI Surveillance**	Ongoing network-wide RF quality surveillance: daily and weekly statistical KPI tracking (Radio Availability, CSSR, CDDR, average throughput, PRB utilization, cell-edge speeds); identification and root-cause analysis of Worst Performing Cells (WPC); dynamic parameter fine-tuning; and weekly optimization meetings. • **Inclusions:** Dedicated RF optimization engineers, weekly WPC triage, parameter tuning scripts, KPI governance. • **Exclusions:** Major network replanning requiring addition of new cell sites (expansion capex).	Per Site / Month	1,000	5,000	12 Months / Year	[Bidder Quote]	Lot 6 Managed Services (Retainer)
MS-RNO-02	**Interference Hunting, PIM Mitigation & Customer Complaint Field Audit**	Field RF investigation for degraded sectors: spectrum analyzer sweeps to detect and locate external interference sources (jamming, unauthorized boosters, microwave leakage); Passive Intermodulation (PIM) testing and remediation on antenna lines; and on-site drive validation for VIP or high-churn barangay clusters. • **Inclusions:** Spectrum analyzer field sweeps, PIM testing, interference source triangulation, customer complaint verification. • **Exclusions:** Legal action against illegal booster operators (handled by NTC / Operator Legal).	Per Site / Month	1,000	5,000	12 Months / Year	[Bidder Quote]	Lot 6 Managed Services (Retainer)

Item Ref	Service Title	Detailed Technical Scope & Boundaries	UoM	Ph 1 Qty	Prog Qty	Evaluation Basis	Unit Rate	Lot / TCO Mapping
MS-FAC-01	**Passive Tower & Civil Facility Maintenance**	Structural preservation of passive site assets: rust scraping, cold galvanizing compound application, and repainting of steel members in corrosive/coastal environments; repairing cracked equipment plinths; re-caulking cable entry boots; and maintaining access road drainage and clear-way. • **Inclusions:** Anti-corrosion touch-up, cold galvanizing, plinth crack sealing, cable entry maintenance. • **Exclusions:** Full tower structural re-erection post-catastrophic failure.	Per Site / Month	1,000	5,000	12 Months / Year	[Bidder Quote]	Lot 6 Managed Services (Retainer)
MS-PWR-01	**Energy, Utility Grid (Meralco/Coop) Coordination & Genset PM**	Power management coordination: monitoring utility power reliability, validating commercial meter readings, coordinating with Meralco/electric cooperatives for transformer fuse restoration, and performing routine 250-hour preventive servicing on backup diesel generator sets (engine oil, oil filter, fuel filter, air cleaner replacement, coolant check). • **Inclusions:** Utility outage coordination, meter reading audit, 250-hr generator PM servicing, coolant/filter checks. • **Exclusions:** Commercial grid electric utility consumption charges (Operator pays utility bills).	Per Site / Month	1,000	5,000	12 Months / Year	[Bidder Quote]	Lot 6 Managed Services (Retainer)
MS-SW-01	**Software Support, Firmware Patching & Cybersecurity Hardening**	Scheduled execution of software maintenance: applying OEM-approved firmware updates and feature patches during maintenance windows (00:00–05:00); implementing CVE security patches and hardening configs; executing weekly automated BBU and Core configuration backups; and maintaining offsite configuration archives. • **Inclusions:** Firmware deployment, CVE security updates, maintenance window execution, automated backups. • **Exclusions:** New major software feature releases requiring feature license fee purchase (Lot 1/Lot 3 scope).	Per Site / Month	1,000	5,000	12 Months / Year	[Bidder Quote]	Lot 6 Managed Services (Retainer)
DOMAIN: AD-HOC CALL-OUT SERVICES RATE CARD (AD-HOC SPECIALIZED SERVICES — ANNUALIZED CONTINGENCY BASKET)								
CALL-01	**Specialized Tower Rigging Call-Out Service**	Ad-hoc mobilization of a certified tower climbing crew to execute non-standard physical antenna modifications: adjusting mechanical tilt, azimuth, or antenna mounting height upon structural reconfiguration or post-typhoon realignment. • **Inclusions:** Rigging crew, climbing tackle, digital compass/inclinometer alignment, before/after photos. • **Exclusions:** Supply of new replacement antennas.	Per Sector	100	500	Contingency Annual Basket	[Bidder Quote]	Ad-Hoc Contingency Rate Card

Item Ref	Service Title	Detailed Technical Scope & Boundaries	UoM	Ph 1 Qty	Prog Qty	Evaluation Basis	Unit Rate	Lot / TCO Mapping
CALL-02	**Emergency Typhoon Disaster Recovery Crew**	Rapid deployment of a specialized 4-man disaster recovery crew equipped with heavy-duty 4x4 vehicle, chainsaws, winches, heavy rigging tackle, and portable generator to clear fallen trees, re-erect toppled poles, and execute emergency structural stabilization post-typhoon. • **Inclusions:** 4-man emergency crew, 4x4 vehicle, chainsaw, winch, generator, debris clearing gear. • **Exclusions:** Supply of complete replacement steel towers.	Per Crew / Day	50	250	Contingency Annual Basket	[Bidder Quote]	Ad-Hoc Contingency Rate Card
CALL-03	**Mobile Emergency Generator Deployment & Continuous Run**	Delivery, positioning, cabling, and commissioning of a mobile trailer-mounted 15–20 kVA diesel generator at a damaged site during extended utility grid blackouts; includes 48 hours of continuous operation, continuous refueling, and operator surveillance. • **Inclusions:** Trailer genset delivery, connection cables, 48 hours continuous running, fuel, technician operator. • **Exclusions:** Operation beyond 48 hours (billed as consecutive additional 48-hr call-out blocks).	Per Deployment (48-hr block)	50	250	Contingency Annual Basket	[Bidder Quote]	Ad-Hoc Contingency Rate Card
CALL-04	**Long-Span Optical Fiber Cable Restoration & Splicing**	Emergency or planned restoration of major backhaul fiber cable cuts exceeding 500 meters: stringing replacement 24-core armored aerial or underground fiber cable, installing suspension clamps, fusion splicing at both splice closures, and testing optical loss. • **Inclusions:** Cable stringing labor, suspension clamps, 2x fusion splice enclosures, OTDR testing. • **Exclusions:** Supply of raw fiber optic cable (provided from operator inventory or billed separately).	Per Span	20	100	Contingency Annual Basket	[Bidder Quote]	Ad-Hoc Contingency Rate Card
CALL-05	**Drone Structural, Panoramic & Thermal Imaging Survey**	High-resolution aerial inspection utilizing drone equipped with 4K optical zoom and radiometric thermal sensor: auditing tower structural integrity, inspecting hard-to-reach antenna brackets, and detecting hotspot anomalies on power joints and cables. • **Inclusions:** Licensed drone pilot, 4K video, high-res photos, radiometric thermal imaging, analysis report. • **Exclusions:** Physical repairs to identified anomalies.	Per Site Flight	50	250	Contingency Annual Basket	[Bidder Quote]	Ad-Hoc Contingency Rate Card

Item Ref	Service Title	Detailed Technical Scope & Boundaries	UoM	Ph 1 Qty	Prog Qty	Evaluation Basis	Unit Rate	Lot / TCO Mapping
CALL-06	**Tower Structural Strengthening & Leg Reinforcement**	Physical structural modification of existing tower or monopole: installing steel reinforcement members, flange stiffeners, cross-bracing, or guy wire tensioning based on approved PE structural engineering calculations. • **Inclusions:** Steel fabrication, rigging installation, bolt torquing, cold galvanizing, PE sign-off. • **Exclusions:** Supply of primary raw steel members (billed based on approved DED BoQ).	Per Tower / Lump Sum based on DED	20	100	Contingency Annual Basket	[Bidder Quote]	Ad-Hoc Contingency Rate Card
CALL-07	**Specialized PIM Hunting & Multi-Sector Interference Remediation**	Advanced on-site PIM testing utilizing specialized 2x20W PIM analyzer: identifying internal and external PIM sources, replacing corroded RF jumpers/connectors, and applying RF shielding tape/materials to achieve PIM <= -155 dBc. • **Inclusions:** Calibrated 2x20W PIM test gear, RF sweep, connector re-termination, shielding materials, sweep report. • **Exclusions:** Replacement of damaged antenna internal phase shifters (OEM warranty).	Per Site / Day	30	150	Contingency Annual Basket	[Bidder Quote]	Ad-Hoc Contingency Rate Card
CALL-08	**Ad-Hoc Emergency Fuel Delivery & Mobile Refueling**	Rapid procurement and transport of clean Euro-4 automotive diesel fuel in certified fuel containers to remote sites during prolonged grid emergencies, including manual transfer to day tank. • **Inclusions:** Fuel procurement, dedicated transport, manual pumping into tank, fuel quality test. • **Exclusions:** Cost of bulk commercial diesel fuel (reimbursed at actual pump receipt price).	Per Liter Delivered	20,000	100,000	Contingency Annual Basket	[Bidder Quote]	Ad-Hoc Contingency Rate Card
SUBTOTAL A1: STANDBY SAQ CONTINGENCY EVALUATION BASKET (103 SITES PH1 / 515 PROG)				**103 Ph1 / 515 Prog**		**1x Contingency**	=SUM(SAQ)	**Lot 5 Standby Rate Card**
SUBTOTAL A2: CORE TURNKEY ROLLOUT SERVICES (1,000 SITES PH1 / 5,000 PROG)				**1,000 Ph1 / 5,000 Prog**		**1x Turnkey**	=SUM(Rollout)	**Lot 5 Core Deployment Baseline**
SUBTOTAL A: TOTAL TURNKEY ROLLOUT SERVICES (LOT 5 RECONCILED BASELINE) [USD]				**Evaluated Baseline**		**Rollout Total**	=A1 + A2	**Reconciles to Lot 5 / Form C Row 6**
SUBTOTAL B1: ANNUALIZED 24/7 MANAGED O&M SERVICES (12-MONTH EVALUATION) [USD/YR]				**1,000 Ph1 / 5,000 Prog**		**Annualized (12 Mo)**	=SUM(O&M)	**Annual Managed O&M Baseline**

Item Ref	Service Title	Detailed Technical Scope & Boundaries	UoM	Ph 1 Qty	Prog Qty	Evaluation Basis	Unit Rate	Lot / TCO Mapping
SUBTOTAL B2: 3-YEAR MANAGED O&M CONTRACTUAL BASELINE (36 MONTHS — LOT 6) [USD]				1,000 Ph1 / 5,000 Prog		36 Months Total	`=B1 × 3`	**Reconciles to Lot 6 / Form C Row 11**
SUBTOTAL C: AD-HOC CALL-OUT CONTINGENCY EVALUATION BASKET (ANNUAL CALL-OFF) [USD]				Nominal Basket		Annual Basket	`=SUM(Call-Out)`	**Ad-Hoc Ceiling Rate Card**
GRAND TOTAL: RECONCILED EVALUATED SERVICES TCO (ROLLOUT + 3-YR O&M + CALL-OUT) [USD]				**Evaluated TCO**		**Master Score**	`=A + B2 + C`	**Official Evaluated Tender TCO**
GRAND TOTAL EQUIVALENT IN PHILIPPINE PESO (PHP @ 55.00 FX BASELINE)				**Evaluated TCO**		**PHP Conversion**	`=USD × 55`	**Official Contract FX Conversion**

4.2 Formal Lot & Form C (10-Year TCO) Reconciliation Mapping Matrix

RFP Commercial Lot / TCO Line Item	Services SOW & Rate Card Mapping	Evaluation / Formula Link	Contractual & Governance Treatment
Lot 5: Site Acquisition, TSSR & Civils (Form C Row 6 Component)	Reconciles directly with **SUBTOTAL A**: • Subtotal A1: Standby SAQ Basket (103 Sites) • Subtotal A2: Core Turnkey Rollout (1,000 Sites)	`=Subtotal A (USD)`	Provides transparent itemized rate card for Lot 5. In baseline tender, bidders quote turnkey lump-sum on Sheet 'Lot 5'. Category A establishes governing unit rates for additions, deductions, and scope variations.
Lot 6: Managed Services & 24/7 Operations (Form C Row 11: Annual O&M Yrs 1-3)	Reconciles directly with **SUBTOTAL B1 & B2**: • Subtotal B1: Annualized O&M (1 Year) • Subtotal B2: 3-Year Contractual Baseline (3 Years)	Annual: `=Subtotal B1` 3-Year: `=Subtotal B2`	Provides detailed module-level rate card for Lot 6 (NOC, FLM, PM Modules A-E, SPMS, RNO). Form C Row 11 applies the annualized rate across 1,000 sites. Retainer-included items evaluate at $0.00 to prevent double-charging.
Contingency Standby SAQ Rate Card (Tier 3 Operator Own-Build Basket)	Items SAQ-01 through SAQ-08 (Nominal 103 Sites / 10.3% Own-Build Basket)	`=Subtotal A1 (USD)`	STANDBY RATE CARD ONLY. Operational baseline is TowerCo-acquired under MLA (897 Co-Loc / 103 B2S). Rates are billable strictly upon written Work Order for authorized Operator Self-Acquired sites.
Ad-Hoc Specialized Services Rate Card (Master Agreement Schedule of Rates)	Items CALL-01 through CALL-08 (Typhoon Crew, Genset, Fiber Splicing, Drone, Fuel)	`=Subtotal C (USD)`	Ceiling rate card for unbudgeted emergency restoration, force majeure events, and specialized engineering call-offs. Billable strictly per approved mobilization event or verified consumption.

5. SERVICE LEVEL AGREEMENTS (SLAS) & LIQUIDATED REMEDIES

Domain	Metric Identifier	Mandatory SLA Target	Operating Floor	Service Credit / Liquidated Remedy
Rollout Delivery	Turnkey Milestone Schedule	**1,000 sites in 365 Days**	Milestone baseline	0.5% milestone contract value per week delay (cap 10%)
Radio Availability	KPI-RAN-01 Network Uptime	**≥ 99.95% Availability**	99.70% Minimum	1.0% monthly O&M fee deduction per 0.1% shortfall
Call Accessibility	KPI-RAN-02 5G Session CSSR	**≥ 97.00% Success**	96.00% Minimum	0.5% monthly O&M fee deduction per 0.2% shortfall below 96.00%
Call Retainability	KPI-RAN-03 Session Drop CDDR	**≤ 1.50% Drops**	2.00% Maximum	0.5% monthly O&M fee deduction per 0.1% excess above 2.00%
Cluster Throughput	KPI-RAN-04 Loaded Sector DL	**≥ 450 Mbps Average**	380 Mbps Floor	Mandatory RF re-tuning & re-drive within 5 days
Cell-Edge Speed	KPI-RAN-05 Cell-Edge DL Speed	**≥ 50 Mbps DL (95% pts)**	25 Mbps Floor	Mandatory eRET electrical tilt re-optimization
FLM Restoration	MTTR Severity 1 (Urban)	**≤ 2.0 Hours MTTR**	2.0 Hours	0.5% monthly site fee per hour delay beyond MTTR
FLM Restoration	MTTR Severity 1 (Rural)	**≤ 4.0 Hours MTTR**	4.0 Hours	0.25% monthly site fee per hour delay beyond MTTR
Preventive Maint.	Quarterly PM Completion	**100% Completed p.a.**	98.0% Floor	Uncompleted site PMs debited 200% of line item rate
Spare Parts Logistics	Advance Replacement (AHR)	**Next Business Day (NBD)**	NBD Standard	$250 USD penalty per calendar day delay in AHR delivery